Appendix III. Papua New Guinea: Debt Sustainability Analysis Update[1]

This debt sustainability analysis (DSA) indicates that Papua New Guinea remains at low risk of debt distress. Baseline projections suggest that PNG's overall public debt is sustainable. Scenario analysis highlights the need to exercise fiscal discipline and to control broad public liabilities—including contingent liabilities—as well as the importance of successfully completing the LNG project to maintain PNG's debt sustainability.

A. Background

PNG has been reducing its public and external debt burden over the past decade.[2] Public sector debt declined from 62 percent of GDP at end-2004 to about 27 percent of GDP at end-2012 (Table 1a). Public and publicly guaranteed (PPG) external debt has also declined sharply, from a peak of over 50 percent of GDP in 2001 to the current 7½ percent of GDP (Table 3a). Around 70 percent of current public external debt is owed to the Asian Development Bank and the World Bank, with the rest mainly to China and Japan.

B. Economic Outlook and Underlying DSA Assumptions

PNG's growth prospects and current account developments over the medium term will be heavily influenced by its extractive sector. Box 1 summarizes the medium-term macroeconomic framework underlying this DSA update. Economic activity is projected to slow significantly in 2013, as the construction phase of the LNG project draws to a close. When LNG production starts in the second half of 2014, real GDP growth is expected to rise to more than 6 percent in 2014 and 21 percent in 2015 when LNG production reaches full capacity. The current account deficit has widened sharply in recent years, largely as a result of increased imports and income outflows related to the LNG project, which are funded by corresponding FDI and medium and long-term loans. The deficit is expected to narrow significantly in 2013, and with LNG exports coming on stream next year, it is set to decline quickly and turn into a surplus in 2015.

[1] As Papua New Guinea is an IBRD/IDA blend country, this DSA update is prepared by Fund staff in close consultation with the World Bank and the Asian Development Bank (AsDB) using the IMF-World Bank DSA framework for Low-Income Countries. The fiscal year for Papua New Guinea is the calendar year. An IMF country team visited Papua New Guinea for the 2013 Article IV consultation from late August to early September 2013.

[2] Papua New Guinea is rated as a medium performer for its policies and institutions for the purposes of the IMF-World Bank low-income country DSA framework. Consequently, the applicable thresholds for PNG for external public debt are: 40 percent for the present value (PV) of the debt-to-GDP ratio; 150 percent for the PV of the debt-to-exports ratio; 250 percent for the PV of the debt-to-revenue ratio; 20 percent for the debt service-to-exports ratio; and 20 percent for the debt service-to-revenue ratio.

C. External Debt Sustainability Analysis

The baseline scenario indicates that all PPG external debt ratios stay well below the indicative thresholds. The external debt burden registers a small increase in the medium term, peaking at around 10 percent before declining in the long run (Figure A1; Table 3). Since additional financing needs for the LNG project and the current account deficit would be met with projected large FDI inflows and medium- and long-term debt drawdowns, the need for public external borrowing is relatively small. PNG's public external debt service ratios are also very low, reflecting its relatively small debt stock as well as the fact that most of its public external debt is highly concessional.

PNG is vulnerable to shocks under the historical scenario despite the current low debt burden, but this scenario has a low probability of materializing. There is a protracted breach of the PV of debt-to-GDP ratio under the historical scenario (Figure A1-b), which could be thought of as circumstances where the LNG project has completely failed or suffered a major delay. In this scenario, when the current account deficit is fixed at the ten-year average of 2003-12, the simulation effectively keeps imports at levels elevated by the LNG project and rules out the expected large increases in LNG exports over time. Given that the LNG project is over 90 percent complete, this scenario represents an unlikely event.

D. Public Debt Sustainability Analysis[3]

The overall public debt dynamics for PNG is favorable, but there are risks. The public debt burden is expected to decline continuously over the projection period under the baseline (Figure A2; Table 1). However, the projected debt path is particularly sensitive to assumptions of GDP growth and fiscal policy. For example, if the expected primary fiscal deficit in 2013 (a year of large fiscal expansion) is assumed to be maintained for the projection period, the PV of the public debt-to-GDP ratio would be above 100 percent by 2030. A scenario to analyze the debt dynamics implied by the authorities' recently released 2014 Budget Strategy Paper (BSP) is also considered. This BSP scenario assumes that the unspent portion of expenditure from 2013 will be rolled over to 2014 and spent. The analysis shows that gross government debt could rise to 37 percent of GDP in 2014, exceeding the 35 percent ceiling set by the government for that year.[4]

Other country-specific liabilities increase risks to debt sustainability. The unfunded superannuation liabilities, estimated to be about 9 percent of GDP at end-2012, remain sizable despite the recent partial payment. Contingent liabilities related to the LNG project are estimated to be about 15 percent of GDP and public enterprise liabilities are about 7½ of GDP. Should these contingent liabilities materialize, there would be a significant increase in the public debt burden over the medium term (Figure 2).

[3] Public debt includes domestic central government debt and external public and publicly guaranteed debt.

[4] The authorities' own projections suggest that the 35 percent of GDP debt ceiling will be observed.

E. Conclusion

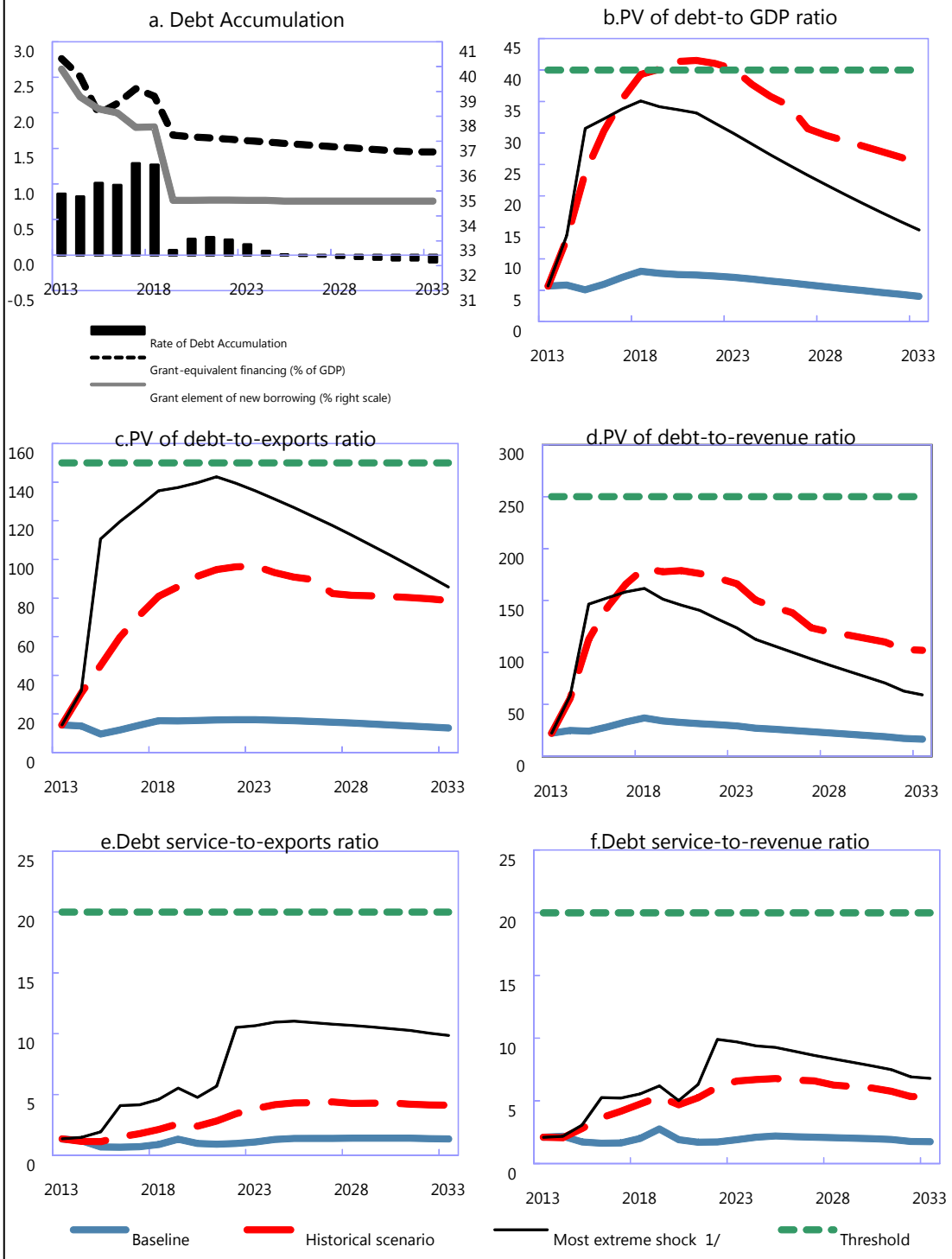
Papua New Guinea's public external debt remains at low risk of debt stress. The baseline scenario indicates that all external PPG debt burden indicators are well below their indicative thresholds. While PNG's external debt dynamics is vulnerable under the historical scenario, the increased debt burden simulated in this scenario is distorted by sharp increases in the current account deficit in recent years related to the LNG project, and thus unlikely to materialize. On the broader public debt, while the government debt burden is relatively low, the existence of large contingent liabilities has significantly increased PNG's debt vulnerability.

This DSA highlights the importance of pursuing a prudent fiscal policy and the need to control broad public liabilities. Staff encourages the authorities to adhere to the government's debt target while focusing on improving spending quality to overcome constrained fiscal space in meeting the country's development needs. Continued structural reforms to increase PNG's growth potential and diversify the economy will also help reduce its vulnerability to external shocks.

BOX 1. MACROECONOMIC ASSUMPTIONS UNDERLYING THE DSA UPDATE

- **Real GDP growth** is projected at 7.6 percent on average over the medium term, and to slow to 3.5 percent in the long run.
- **Inflation** is expected to stabilize over the medium term at about 6 percent and is maintained at this level in the long run.
- **The current account** (including grants) will remain in deficit until 2014, primarily reflecting the strong import growth related to the construction phase of the LNG project. It will turn into a surplus in 2015 as LNG production comes on stream and imports related to the LNG project subside, and is projected to be 7 percent of GDP on average during 2015-25.
- **The grant element of loans** is expected to decline. As per capita income rises, the share of external financing provided on concessional terms is expected to decline gradually over the projection period.
- **The primary fiscal balance** is estimated to be in deficit of 5.2 percent of GDP in 2013. During the current medium-term fiscal strategy period (2013-2017), a continuous primary deficit of 3 percent of GDP on average is expected. After 2023, the primary fiscal balance is projected to turn into surplus, with an average of 0.5 percent of GDP during 2023-33.

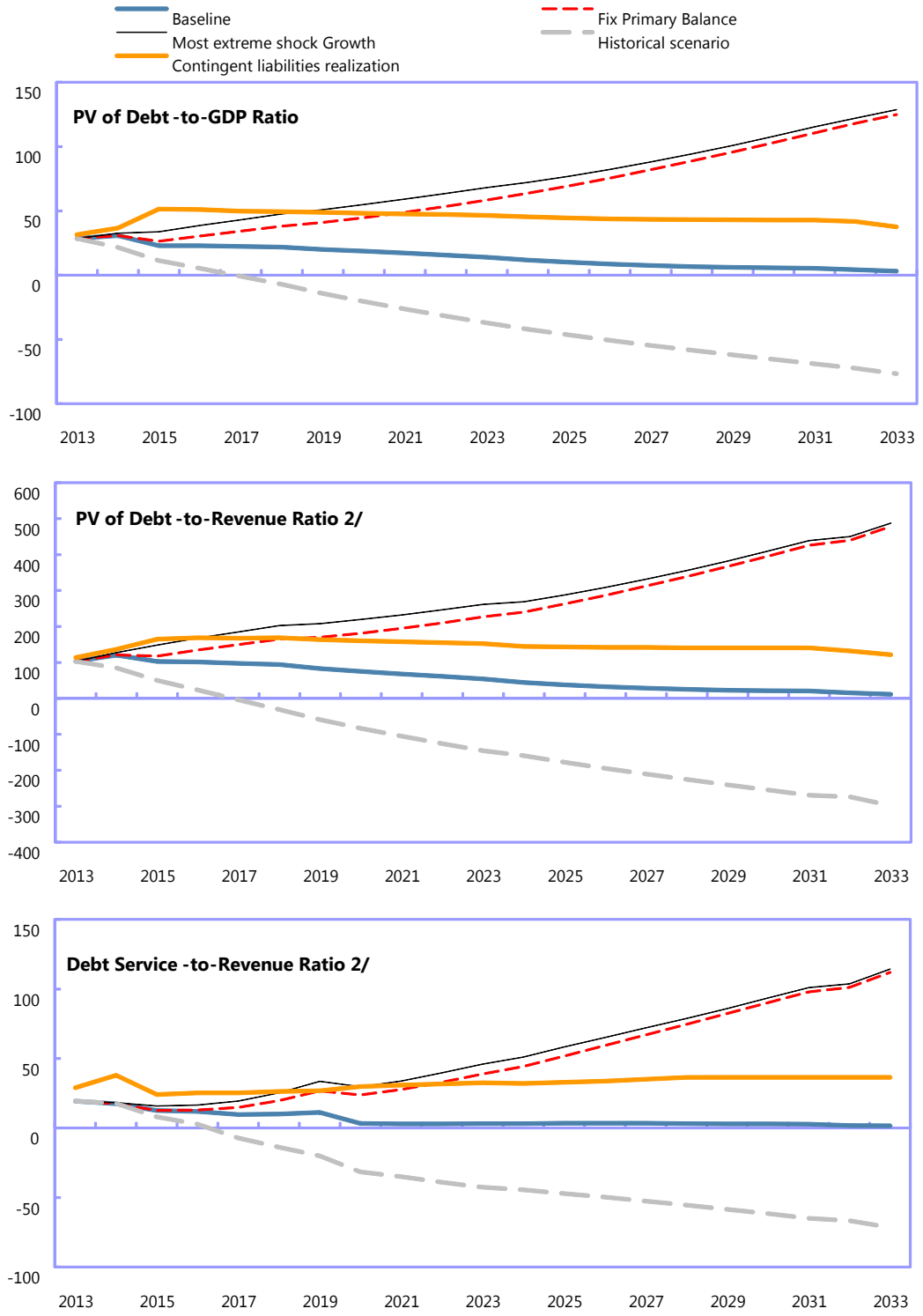
Figure A1. Papua New Guinea: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2013-2033 1/



Sources: Papua New Guinea authorities; and IMF staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2023. In figure b. it corresponds to a Combination shock; in c. to an Exports shock; in d. to a Combination shock; in e. to an Exports shock and in figure f. to a Combination shock

Figure A2.Papua New Guinea: Indicators of Public Debt Under Alternative Scenarios, 2013-2033 1/



Sources: Papua New Guinea authorities; and IMF staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2023.
2/ Revenues are defined inclusive of grants.

Table 1.Papua New Guinea: Public Sector Debt Sustainability Framework, Baseline Scenario, 2010-2033

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2010	2011	2012	Average 1/	Standard Deviation 1/	2013	2014	2015	2016	2017	2018	2013-18 Average	2023	2033	2019-33 Average
Public sector debt 2/	25.3	22.3	26.7			30.9	32.6	24.7	25.1	24.7	24.3		13.6	-0.5	
of which: foreign-currency denominated	10.1	6.8	7.4			8.1	7.8	6.9	8.1	9.5	10.5		6.8	0.5	
Change in public sector debt	-6.0	-3.0	4.4			4.1	1.7	-7.9	0.4	-0.3	-0.5		-2.0	-1.0	
Identified debt-creating flows	-8.2	-6.7	2.0			4.7	1.8	-6.8	1.0	0.3	0.3		-0.7	0.0	
Primary deficit	-4.2	-2.5	1.9	-3.7	5.6	5.3	5.2	2.0	1.4	1.2	1.2	2.7	-0.2	-0.1	-0.3
Revenue and grants	31.3	30.4	29.6			27.6	25.4	22.4	22.6	22.9	23.1		25.7	26.1	
of which: grants	5.3	3.4	3.4			2.0	1.9	1.4	1.5	1.5	1.4		1.5	1.3	
Primary (noninterest) expenditure	27.1	27.9	31.5			32.9	30.7	24.4	24.0	24.1	24.3		25.5	25.9	
Automatic debt dynamics	-4.0	-4.2	0.1			-0.6	-3.5	-8.8	-0.5	-0.8	-0.9		-0.5	0.1	
Contribution from interest rate/growth differential	-2.9	-2.4	-0.1			-0.9	-3.2	-8.3	-0.7	-1.0	-1.0		-0.5	0.1	
of which: contribution from average real interest rate	-0.6	0.1	1.6			0.5	-1.4	-2.5	0.1	-0.2	-0.2		0.0	0.1	
of which: contribution from real GDP growth	-2.2	-2.4	-1.7			-1.4	-1.8	-5.7	-0.8	-0.8	-0.8		-0.5	0.0	
Contribution from real exchange rate depreciation	-1.1	-1.9	0.2			0.3	-0.2	-0.6	0.2	0.2	0.1		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	2.2	3.7	2.5			-0.5	-0.1	-1.0	-0.6	-0.7	-0.7		-1.3	-0.9	
Other Sustainability Indicators															
PV of public sector debt	...	...	24.3			28.5	30.6	22.9	22.9	22.4	21.8		13.9	3.1	
of which: foreign-currency denominated	...	...	5.1			5.7	5.8	5.1	6.0	7.1	8.0		7.1	4.1	
of which: external	...	...	5.1			5.7	5.8	5.1	6.0	7.1	8.0		7.1	4.1	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 3/	6.3	6.0	12.3			18.1	17.5	11.4	10.3	9.2	8.9		0.6	0.3	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	82.4			103.1	120.5	102.5	101.3	97.6	94.1		54.3	11.8	
PV of public sector debt-to-revenue ratio (in percent)	...	...	93.0			111.3	130.2	109.5	108.4	104.7	100.4		57.6	12.5	
of which: external 4/	...	...	19.3			22.2	24.8	24.2	28.2	33.1	37.0		29.2	16.4	
Debt service-to-revenue and grants ratio (in percent) 5/	14.8	15.2	19.3			19.4	17.6	12.6	11.9	9.8	10.0		3.0	1.7	
Debt service-to-revenue ratio (in percent) 5/	17.8	17.1	21.8			21.0	19.0	13.5	12.7	10.5	10.7		3.2	1.8	
Primary deficit that stabilizes the debt-to-GDP ratio	1.9	0.5	-2.5			1.2	3.5	9.9	1.0	1.5	1.7		1.8	0.8	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	7.7	10.7	8.1	5.8	3.0	5.4	6.3	21.4	3.3	3.4	3.5	7.2	3.0	3.0	3.1
Average nominal interest rate on forex debt (in percent)	1.6	1.5	1.7	2.4	0.7	1.5	1.7	1.9	1.8	1.7	1.7	1.7	2.2	16.8	6.1
Average real interest rate on domestic debt (in percent)	-3.9	1.0	11.0	5.2	5.4	2.9	-6.5	-12.3	0.7	-1.1	-1.2	-2.9	-0.7	...	9.1
Real exchange rate depreciation (in percent, + indicates depreciation)	-10.0	-20.6	2.7	-7.9	8.4	3.7	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	9.9	4.3	-2.6	4.6	4.6	2.2	9.4	16.1	2.7	3.5	3.8	6.3	5.0	5.0	5.1
Growth of real primary spending (deflated by GDP deflator, in percent)	-0.2	0.1	0.2	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	...	...	39.9	38.8	38.3	38.1	37.6	37.6	38.4	34.6	34.6	...

Sources: Papua New Guinea authorities; and IMF staff estimates and projections.

1/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

2/ Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.

3/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

4/ Revenues excluding grants.

5/ Debt service is defined as the sum of interest and amortization of medium- and long-term debt.

Table 2.Papua New Guinea: Sensitivity Analysis for Key Indicators of Public Debt 2013-2033

(In percent)

	Projections							
	2013	2014	2015	2016	2017	2018	2023	2033
PV of Debt-to-GDP Ratio								
Baseline	28	31	23	23	22	22	14	3
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	28	22	11	5	-1	-7	-37	-77
A2. Primary balance is unchanged from 2013	28	31	26	30	34	38	58	125
A3. Permanently lower GDP growth 1/	28	31	24	24	24	25	24	44
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviation in 2014-2015	28	33	34	39	43	47	68	129
B2. Primary balance is at historical average minus one standard deviation in 2014-2015	28	27	20	20	19	19	10	-1
B3. Combination of B1-B2 using one half standard deviation shocks	28	25	17	21	25	28	43	88
B4. One-time 30 percent real depreciation in 2014	28	33	25	25	24	23	17	9
B5. 10 percent of GDP increase in other debt-creating flows in 2014	28	41	32	32	32	31	24	16
PV of Debt-to-Revenue Ratio 2/								
Baseline	103	121	102	101	98	94	54	12
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	103	85	50	23	-4	-31	-145	-298
A2. Primary balance is unchanged from 2013	103	121	118	134	150	165	227	478
A3. Permanently lower GDP growth 1/	103	122	105	106	106	106	92	168
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviation in 2014-2015	103	128	148	168	185	203	262	488
B2. Primary balance is at historical average minus one standard deviation in 2014-2015	103	107	89	87	84	80	40	-5
B3. Combination of B1-B2 using one half standard deviation shocks	103	98	77	93	106	120	164	335
B4. One-time 30 percent real depreciation in 2014	103	130	110	109	105	101	68	36
B5. 10 percent of GDP increase in other debt-creating flows in 2014	103	161	142	141	138	135	94	60
Debt Service-to-Revenue Ratio 2/								
Baseline	19	18	13	12	10	10	3	2
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	19	18	8	3	-7	-14	-43	-72
A2. Primary balance is unchanged from 2013	19	18	13	13	15	20	39	112
A3. Permanently lower GDP growth 1/	19	18	13	12	11	11	10	36
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviation in 2014-2015	19	18	16	17	19	25	46	114
B2. Primary balance is at historical average minus one standard deviation in 2014-2015	19	18	11	9	6	6	-1	-3
B3. Combination of B1-B2 using one half standard deviation shocks	19	18	10	7	2	5	22	77
B4. One-time 30 percent real depreciation in 2014	19	18	13	13	11	11	6	7
B5. 10 percent of GDP increase in other debt-creating flows in 2014	19	18	19	21	20	21	13	14

Sources: Papua New Guinea authorities; and IMF staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Table 3a.: External Debt Sustainability Framework, Baseline Scenario, 2010-2033 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 2/	Standard Deviation 2/	Projections									
	2010	2011	2012			2013	2014	2015	2016	2017	2018	2013-2018 Average	2023	2033	2019-2033 Average
External debt (nominal) 1/	**71.0**	**86.5**	**129.1**			**120.2**	**101.8**	**68.8**	**61.5**	**53.2**	**48.6**		**20.3**	**0.6**	
of which: public and publicly guaranteed (PPG)	10.1	6.8	7.4			8.1	7.8	6.9	8.1	9.5	10.5		6.8	0.5	
Change in external debt	34.8	15.5	42.6			-8.9	-18.4	-33.0	-7.3	-8.3	-4.6		-4.7	-0.1	
Identified net debt-creating flows	9.7	-1.9	25.6			14.8	-16.4	-31.0	-14.1	-13.4	-12.2		-8.2	-3.2	
Non-interest current account deficit	**19.5**	**22.5**	**49.5**	**7.0**	**19.5**	**25.7**	**0.8**	**-12.9**	**-9.7**	**-9.2**	**-8.2**		**-5.5**	**-1.5**	-3.8
Deficit in balance of goods and services	12.4	21.1	35.9			14.7	-2.4	-21.5	-20.8	-20.3	-19.2		-15.0	-10.2	
Exports	63.5	56.5	43.5			39.7	42.5	52.4	50.8	49.9	48.6		41.6	32.1	
Imports	75.8	77.6	79.4			54.4	40.1	30.9	30.0	29.6	29.4		26.6	21.9	
Net current transfers (negative = inflow)	-2.0	-1.9	-1.1	-3.6	2.4	-0.8	-1.1	-0.7	-0.7	-0.7	-0.8		-1.2	0.1	-0.6
of which: official	-2.8	-3.1	-2.8			-2.5	-2.1	-1.5	-1.5	-1.5	-1.5		-1.5	-1.3	
Other current account flows (negative = net inflow)	9.0	3.3	14.7			11.8	4.4	9.3	11.8	11.8	11.8		10.7	8.6	
Net FDI (negative = inflow)	**-5.7**	**-8.0**	**-12.4**	**-5.8**	**5.6**	**-5.6**	**-11.6**	**-2.6**	**-2.6**	**-2.6**	**-2.5**		**-2.2**	**-1.8**	-2.1
Endogenous debt dynamics 3/	**-4.1**	**-16.4**	**-11.4**			**-5.3**	**-5.6**	**-15.5**	**-1.8**	**-1.6**	**-1.4**		**-0.5**	**0.1**	
Contribution from nominal interest rate	1.9	1.0	1.5			1.2	1.1	0.6	0.4	0.4	0.3		0.2	0.1	
Contribution from real GDP growth	-2.3	-5.7	-6.0			-6.6	-6.8	-16.1	-2.2	-2.0	-1.8		-0.7	0.0	
Contribution from price and exchange rate changes	-3.7	-11.7	-7.0			...	...	...	...	...	...		...	...	
Residual (3-4) 4/	**25.1**	**17.4**	**17.0**			**-23.8**	**-2.0**	**-2.0**	**6.9**	**5.1**	**7.5**		**3.5**	**3.1**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 5/	...	...	126.7			117.8	99.9	67.0	59.3	50.8	46.1		20.7	4.1	
In percent of exports	...	...	291.5			296.7	234.9	127.7	116.8	101.7	94.8		49.6	12.8	
PV of PPG external debt	**...**	**...**	**5.1**			**5.7**	**5.8**	**5.1**	**6.0**	**7.1**	**8.0**		**7.1**	**4.1**	
In percent of exports	**...**	**...**	**11.6**			**14.3**	**13.7**	**9.7**	**11.7**	**14.2**	**16.5**		**17.0**	**12.7**	
In percent of government revenues	**...**	**...**	**19.3**			**22.2**	**24.8**	**24.2**	**28.2**	**33.1**	**37.0**		**29.2**	**16.4**	
Debt service-to-exports ratio (in percent)	**15.5**	**16.2**	**15.4**			**9.0**	**31.6**	**15.8**	**15.7**	**17.5**	**9.9**		**9.2**	**1.5**	
PPG debt service-to-exports ratio (in percent)	**1.4**	**1.2**	**1.4**			**1.4**	**1.2**	**0.7**	**0.7**	**0.7**	**0.9**		**1.1**	**1.4**	
PPG debt service-to-revenue ratio (in percent)	**3.3**	**2.4**	**2.4**			**2.1**	**2.2**	**1.7**	**1.6**	**1.6**	**2.0**		**1.9**	**1.8**	
Total gross financing need (Billions of U.S. dollars)	2.3	3.0	6.6			3.8	0.5	-1.8	-1.1	-0.8	-1.6		-1.3	-1.5	
Non-interest current account deficit that stabilizes debt ratio	-15.3	6.9	6.8			34.7	19.2	20.1	-2.5	-0.9	-3.6		-0.8	-1.4	
Key macroeconomic assumptions															
Real GDP growth (in percent)	7.7	10.7	8.1	5.8	3.0	5.4	6.3	21.4	3.3	3.4	3.5	7.2	3.0	3.0	3.1
GDP deflator in US dollar terms (change in percent)	11.3	19.7	8.8	11.2	6.9	1.2	5.3	11.7	-1.3	-0.5	0.2	2.8	1.9	1.8	1.8
Effective interest rate (percent) 6/	6.4	1.9	2.1	3.2	1.3	1.0	1.1	0.8	0.7	0.7	0.7	0.8	0.9	16.2	5.2
Growth of exports of G&S (US dollar terms, in percent)	30.9	17.9	-9.6	14.5	17.4	-2.6	19.8	67.2	-1.2	1.1	0.9	14.2	2.2	2.3	2.1
Growth of imports of G&S (US dollar terms, in percent)	40.1	35.5	20.3	21.6	13.0	-26.9	-17.5	4.5	-1.1	1.5	3.0	-6.1	3.0	2.8	2.9
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	39.9	38.8	38.3	38.1	37.6	37.6	38.4	34.6	34.6	34.6
Government revenues (excluding grants, in percent of GDP)	26.0	27.1	26.2			25.6	23.5	20.9	21.1	21.4	21.7		24.2	24.7	24.4
Aid flows (in Billions of US dollars) 7/	0.6	0.5	0.8			0.5	0.6	0.6	0.8	0.9	0.9		0.6	0.8	
of which: Grants	0.5	0.4	0.5			0.3	0.3	0.4	0.4	0.4	0.4		0.5	0.7	
of which: Concessional loans	0.1	0.0	0.2			0.2	0.2	0.3	0.4	0.5	0.5		0.1	0.1	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			2.8	2.5	2.0	2.1	2.3	2.2		1.6	1.4	1.6
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			71.7	71.5	68.9	66.7	63.9	62.8		86.1	91.3	87.3
Memorandum items:															
Nominal GDP (Billions of US dollars)	9.7	12.9	15.1			16.1	18.1	24.5	25.0	25.7	26.7		34.1	55.1	
Nominal dollar GDP growth	19.9	32.4	17.6			6.7	11.9	35.6	2.0	2.9	3.7	10.5	4.9	4.9	5.0
PV of PPG external debt (in Billions of US dollars)			0.8			0.9	1.0	1.2	1.5	1.8	2.1		2.4	2.2	
(PVt-PVt-1)/GDPt-1 (in percent)						0.9	0.8	1.0	1.0	1.3	1.3	1.0	0.1	-0.1	0.0
Gross workers' remittances (Billions of US dollars)	-0.1	-0.2	-0.3			-0.3	-0.2	-0.2	-0.2	-0.2	-0.2		-0.1	-0.8	
PV of PPG external debt (in percent of GDP + remittances)	...	...	5.1			5.8	5.9	5.1	6.0	7.1	8.1		7.1	4.1	
PV of PPG external debt (in percent of exports + remittances)	...	...	12.1			15.0	14.1	9.8	11.9	14.4	16.8		17.1	13.3	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	1.5			1.4	1.2	0.7	0.7	0.7	0.9		1.1	1.4	

Sources: Papua New Guinea authorities; and IMF staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

3/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

4/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

5/ Assumes that PV of private sector debt is equivalent to its face value.

6/ Current-year interest payments divided by previous period debt stock.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 3b.Papua New Guinea: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013-2033

(In percent)

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
PV of debt-to GDP ratio								
Baseline	6	6	5	6	7	8	**7**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	6	13	24	30	35	39	**40**	25
A2. New public sector loans on less favorable terms in 2013-2033 2/	6	6	6	7	8	10	**9**	7
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	6	6	6	7	8	10	**9**	5
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	6	11	27	29	30	31	**27**	13
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	6	6	5	6	7	9	**8**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	6	13	12	13	14	15	**13**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	6	14	31	32	34	35	**30**	15
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	6	8	7	8	10	11	**10**	6
PV of debt-to-exports ratio								
Baseline	14	14	10	12	14	17	**17**	13
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	14	31	45	60	71	81	**97**	79
A2. New public sector loans on less favorable terms in 2013-2033 2/	14	14	11	14	17	20	**23**	22
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	14	13	9	11	14	16	**17**	12
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	14	32	111	120	127	136	**136**	86
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	14	13	9	11	14	16	**17**	12
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	14	31	23	25	28	31	**31**	21
B5. Combination of B1-B4 using one-half standard deviation shocks	14	37	87	95	101	108	**107**	68
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	14	13	9	11	14	16	**17**	12
PV of debt-to-revenue ratio								
Baseline	22	25	24	28	33	37	**29**	16
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	22	56	113	143	166	181	**166**	102
A2. New public sector loans on less favorable terms in 2013-2033 2/	22	26	27	33	40	46	**39**	28
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	22	25	29	34	40	45	**35**	20
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	22	47	130	135	140	143	**110**	52
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	22	25	26	30	35	39	**31**	17
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	22	56	57	61	66	69	**53**	27
B5. Combination of B1-B4 using one-half standard deviation shocks	22	58	147	153	158	162	**124**	59
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	22	34	33	39	45	51	**40**	23

Table 3b.Papua New Guinea: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013-2033 (concluded)

(In percent)

Debt service-to-exports ratio								
Baseline	1	1	1	1	1	1	**1**	1
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	1	1	1	2	2	2	**4**	4
A2. New public sector loans on less favorable terms in 2013-2033 2/	1	1	1	1	1	1	**2**	2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	1	1	1	1	1	1	**1**	1
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	1	1	2	4	4	5	**11**	10
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	1	1	1	1	1	1	**1**	1
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	1	1	1	1	1	1	**2**	2
B5. Combination of B1-B4 using one-half standard deviation shocks	1	1	2	3	3	4	**8**	8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	1	1	1	1	1	1	**1**	1
Debt service-to-revenue ratio								
Baseline	2	2	2	2	2	2	**2**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	2	2	3	4	4	5	**7**	5
A2. New public sector loans on less favorable terms in 2013-2033 2/	2	2	2	2	2	2	**3**	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	2	2	2	2	2	2	**2**	2
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	2	2	2	5	5	5	**9**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	2	2	2	2	2	2	**2**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	2	2	2	3	3	3	**4**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	2	2	3	5	5	6	**10**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	2	3	2	2	2	3	**3**	2
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	34	34	34	34	34	34	**34**	34

Sources: Papua New Guinea authorities; and IMF staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), noninterest current account in percent of GDP, and non-debt-creating flows.
2/ Assumes that the interest rate on new borrowing is 2 percentage points higher than in the baseline, while grace and maturity periods are the same as in the baseline.
3/ Export values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.